Lightest full-fidelity feasible design per gas, u_res_max = 0.5 m/s — full run

Horizontal Displacement
x [m]
Horizontal Velocity
u_x [m/s]
Horizontal Acceleration
a_x [m/s^2]
-A_LIM
Horizontal Force
F_x [N]
Vertical Penetration
y_pen [m]
Displacement Along n
Vertical Velocity
u_y [m/s]
Velocity Along n
Vertical Acceleration
a_y [m/s^2]
-A_LIM
Bag Load Along n (F_n/M)
Net Vertical Force (incl. g)
F_y,net [N]
Pressure Force Along n
h [m]
dotted: D0 of each design
u_n [m/s]
F_n/M [m/s^2]
A_LIM
F_n [N]
Gas Mass
m [kg]
Time [s]
Airbag Volume
V [m^3]
Time [s]
1e5
Internal Pressure
P_bag [Pa]
Time [s]
Gas Temperature
T [K]
Time [s]
N2: cylinder / Nylon, m = 0.583 kg
He: cylinder / Nylon, m = 0.379 kg
Ar: cylinder / Nylon, m = 0.732 kg
CO2: cylinder / Nylon, m = 0.707 kg
dry air: cylinder / Nylon, m = 0.591 kg